| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 06/22/2005 |
| **PI Name:** | Caiozzo, Vincent J. Ph.D. | | |
| **Project Title:** | Hypergravity Resistance Training: Countermeasure to Microgravity | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Muscle Alterations and Atrophy Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | vjcaiozz@uci.edu | **Fax:** | FY 949-824-3278 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 949-824-5571 |
| **Organization Name:** | University of California, Irvine | | |
| **PI Address 1:** | Medical Sciences Bldg I, Room B152 | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Irvine | **State:** | CA |
| **Zip Code:** | 92697-0001 | **Congressional District:** | 48 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Biomedical Research & Countermeasures 03-OBPR-04 |
| **Start Date:** | 04/01/2004 | **End Date:** | 03/31/2008 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Baldwin, Kenneth ( University of California, Irvine )<br>Adams, Gregory ( University of California, Irvine )<br>Cavanagh, Peter ( The Cleveland Clinic Foundation )<br>Kreitenberg, Art ( University of California, Irvine ) | | |
| **Grant/Contract No.:** | NCC 9-58-MA00403 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | A program priority of NASA's Biomedical Research and Countermeasures Program (NRA-03-OBPR-04) is to determine the potential usefulness of artificial gravity as a countermeasure, especially with respect to skeletal muscle atrophy and loss of muscle function. The current proposal is a "proof-of-principle" of a unique countermeasure technology referred to as the "Space Cycle." The Space Cycle is a human powered centrifuge that can be used to generate various levels of artificial/hypergravity. Artificial gravity/hypergravity in a microgravity environment could be used as a novel method of performing resistance training under high loading conditions. The novelty of artificial gravity/hypergravity resistance training is that each element of the body is loaded proportional to the local gravitational field, and under hypergravity conditions muscles like those of the leg can be made to work against very high loads (e.g., + 2 body weights) without the need for external weights. The primary objective of this proposal is to use the Space Cycle to address the following general hypothesis: Artificial gravity can be used as a unique resistance training modality that acts as an effective countermeasure, preventing the loss of muscle mass and function that occurs due microgravity. In addressing this issue, a logical sequence of experiments is proposed with the following objectives: i) determine if squats performed under hypergravity conditions and without external weights can produce foot forces similar to those seen when performing squat resistance training (SRT) under normal 1 G conditions; ii) determine if squats performed under hypergravity conditions produce muscle adaptations similar to those seen using a squat resistance training program under normal 1 G conditions; and iii) determine if squat hypergravity resistance training (SHRT) program is an effective countermeasure to simulated microgravity. For the purposes of this grant proposal, we are focusing on SRT because squats recruit a broad spectrum of muscles in the leg and back, and are one of the classical exercises used by bodybuilders and athletes to hypertrophy muscles of the leg. Additionally, the so-called antigravity muscles of the leg are at the greatest risk for atrophy induced by microgravity. Furthermore, squats are a target exercise performed by astronauts on the International Space Station. During the first year of funding, the primary focus was on four goals: 1) organization of the team; 2) modifying the Space Cycle so that squats could be performed under hypergravity conditions; 3) integration of equipment for measuring foot forces, EMG, and joint angle; and 4) IRB modifications. With respect to the first issue, we held a mini retreat in June of 2004 to map out our strategy for the first year. Dr. Cavanagh also attended this meeting. The Space Cycle team will be having another mini retreat March 31, 2005 to begin planning actual experiments. From an organizational perspective, the Space Team has montly teleconference meetings that include all members of the Space Cycle team. Finally, with respect to this first issue, Dr. Cavanagh was successful in finding a postdoctoral fellow, Dr. Mark Pierre, to assist with the biomechanical measurements. The Space Cycle has gone through three iterations of modification. Engineering drawings were first developed, and following this first draft we have made several modifications to the arm that will be used for performing the squat exercise. Some of the key issues that were resolved are related to centrifuge balance issues and moment of inertia issues. The data collection system involves a Konigsberg telemetry system for the collection of EMG data. This system has been integrated with a Pedar telemetry system for the measurement of foot forces while performing squats. Each of these packages is complemented with the measurement of joint angles. The last major objective that was addressed this past year was a modification of IRB protocols. This consumed a significant amount of time given the novelty of the Space Cycle as compared to other more accepted protocols. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | There are potentially several unique aspects of the Space Cycle that may have a direct benefit to earth based activities. Currently, it is unclear whether hypergravity may represent a unique loading mode that might enhance training and/or rehabilitation. The unique aspect of hypergravity is that each element is weighted in accordance to its specific gravitational field. In essence, this is like hanging a set of bar bells on every atom/molecule in the body. This distribution of load may be quite different than that of placing weights on a given body structure whereby there is a concentrated loading on that particular structure. Additionally, hypergravity may represent a unique modality for spinal cord injured patients. It is well known that many of these individuals have a reduced orthostatic tolerance. Some have suggested that exposing such individuals to hypergravity may help in this regard. We hope to explore this avenue of investigation in the coming years. |
| **Task Progress:** | The primary tasks of the past year were to:<br>1. Redesign the Space Cycle so that squat resistance training could be performed. 2. Purchase and develop methods for making measurements of foot forces while performing squats under different hypergravity conditions. 3. Purchase and develop methods for making EMG measurements during Space Cycle activity. 4. Develop methods for making joint angle measurements. 5. Conduct a national search for a postdoctoral fellow to participate in the biomechanical aspects of this study. 6. To recruit individuals to participate in the Space Cycle project. |
| **Bibliography Type:** | Description: (Last Updated: 12/17/2008) |
| **Articles in Peer-reviewed Journals** | Caiozzo VJ, Rose-Gottron C, Baldwin KM, Cooper D, Adams G, Hicks J, Kreitenberg A. "Hemodynamic and metabolic responses to hypergravity on a human-powered centrifuge." Aviat Space Environ Med. 2004 Feb;75(2):101-8. PMID: 14960043 , Feb-2004 |
| **Presentation** | Caiozzo, V. J. "Hypergravity resistance training: countermeasure to microgravity " Jan-2005 |